

NBA opening-spread walk-forward — model and strategy review

Code, data pipeline and the full research register: github.com/sean-qin-usa/nba-prediction-model

Strategy

Directional value-taking against the opening point spread in NBA regular-season games. A market-blind margin model prices every listed game; its disagreement with the opening spread is then spent, at about a third of face value, as a correction to the opening spread — $m_{final} = open_margin + f(x)$, with f a ridge shrunk hard toward zero and fitted walk-forward. A single side is taken when the corrected margin still disagrees with the line by more than a walk-forward selected threshold, at the best number available across books. Every position is one side of one game at -110 , entered at the open and held to settlement: no hedge, no offsetting leg, no in-game management, no exit. Flat 1 unit per bet, so nothing compounds. No calendar filter.

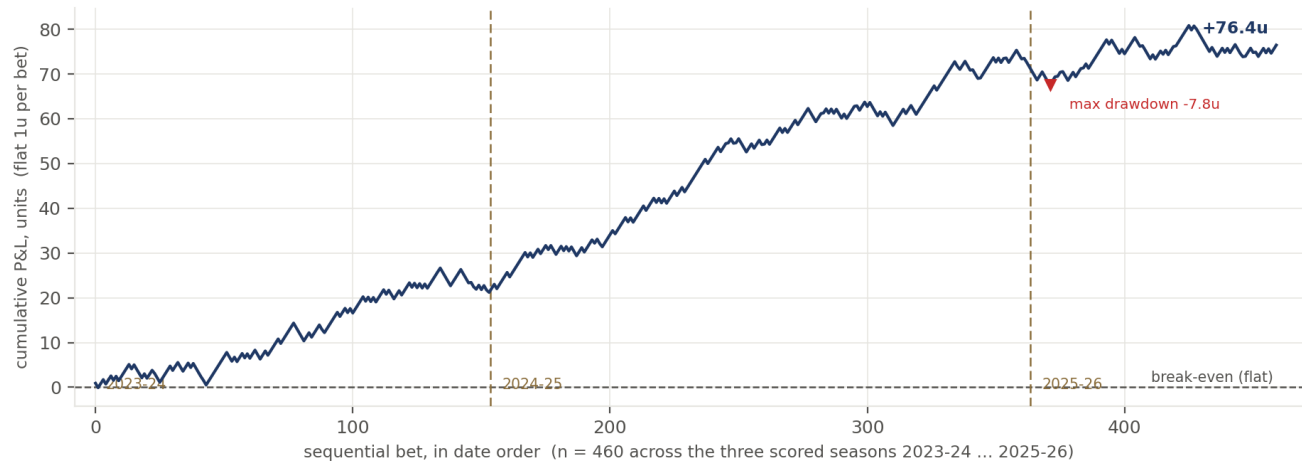
Selection is the strategy. The book below is 460 bets out of roughly 3,690 regular-season games in the window, about 12% of the slate. A side is taken only where the corrected margin disagrees with the opening spread by more than the walk-forward threshold, so the cover rate on the selected book is not the cover rate on the slate.

Simulation: 460 bets, 3 seasons, 2023-24 → 2025-26. The configuration is selected on seasons 1..k from a pre-declared space, frozen, scored on season k+1, and rolled forward; nothing is refitted on the season being scored. The selection history reaches back to 2012-13, so the three seasons here are scored against a decade of prior data rather than against each other. Each bet is priced at the best of the 9 books held, the maximum observed at the open.

Headline results

season	books/game at open	bets	P&L (u)	ROI	Sharpe (ann.)
2023-24	7.74	154	+21.20u	+13.77%	1.90
2024-25	1.00	210	+50.40u	+24.00%	3.81
2025-26	1.03	96	+4.85u	+5.05%	0.54
2023-26 pooled, modelled best of 9 books	—	460	+76.44u	+16.62%	2.25
2019-26, all seven scored seasons	—	888	+80.93u	+9.11%	1.10

ROI is P&L per unit staked; Sharpe is annualized on realised sessions. The multi-book panel is measured only in 2023-24, so the pooled figures are modelled for 2024-25 and 2025-26. The last row is the full scored frame, the honest denominator against which the three-season block is its recent end.



All 460 bets in date order at the opening spread and -110 ; flat 1u, so the path is a running sum and nothing is re-selected within it. Max drawdown is $-7.75u$ at bet 372. Over the full seven-season frame the path is $+80.93u$ on 888 bets with a $-23.1u$ drawdown, the first three of those seasons losing money and the last four making $+83.5u$.

Pooled ROI is $+16.62\%$ over 460 bets, with a season-clustered 95% interval of $[-9.29\%, +37.83\%]$; on the full seven-season frame it is $+9.11\%$ with an interval of $[-2.75\%, +16.08\%]$. Three seasons is a small sample and both intervals contain zero. 2024-25 supplies 66% of the three-season P&L.

The model

The model outputs a margin, not a probability. A spread is itself a margin forecast, so the sides test involves no devig or probability conversion.

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margin = 0.5·four_factors + 0.5·availability_composition + schedule_layer + tank_term
P(home win) = sigmoid(margin / 7.2)
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Two independent estimates of team strength, averaged, plus additive context. The market-blind model never sees market odds, enforced structurally rather than by convention. Only the offset layer above it sees the price.

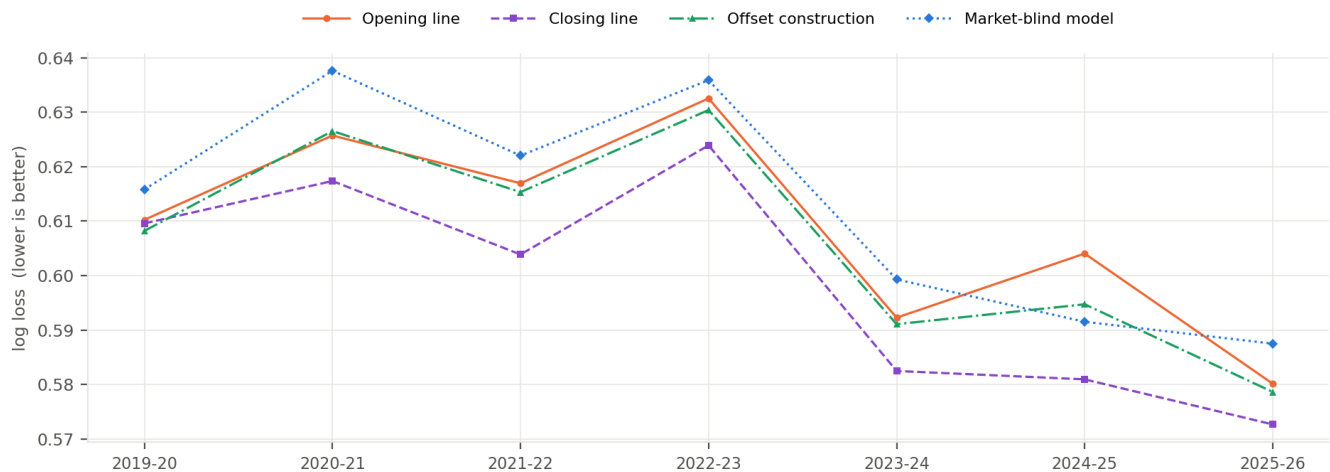
component	what it is	how it is fit
Four factors	opponent-adjusted ratings on shooting, turnovers, rebounding and free-throw rate; a decomposition of points per possession rather than a feature shortlist	one L2 ridge solve per factor (ridge=25), then the four adjusted values mapped to points by a fitted linear map. An 8-factor extension tied, so 4 is kept for parsimony.
Availability composition	Σ over available players of DARKO talent $\times$ trailing minutes / 48; the leg that reacts to injuries	each player weighted by $1 - P(\text{out})$, forecast from information available at the open. Requires the daily injury report; this is why the model frame starts 2019-20
Schedule layer	home edge, back-to-backs, dead-team flags	estimated walk-forward, EB shrinkage $n/(n+600)$ toward a prior, team_home_ridge=200 on per-team home deviations. The only component that has survived strict out-of-sample testing on every split tried.
Tank term	late-season effort	exactly zero outside its window
Offset layer	the correction applied to the opening line	ridge on three features knowable at the open, shrunk hard toward zero. Fitted edge coefficient 0.33–0.37 in every fold

Regularisation, in four forms: L2 ridge on the ratings solve; empirical-Bayes shrinkage toward a prior in the schedule and tank layers; data augmentation via prior-season pseudo-observations; and Bayesian/EB priors in the usage and props fits. The link scale 7.2 is a plug-in, not a hand-set constant: matching a logistic to the margin-residual normal gives 7.53 from training residuals alone.

Inputs are four, and all public: nba_api box scores, the daily NBA injury report (PDF ingest), DARKO talent ratings, and historical odds, the last used by the offset layer and for scoring but never by the market-blind model. Measured: the entire purchasable stack (professional minutes projections, tracking feeds, premium talent ratings) is worth +0.0012 of log loss combined, not significant. Fitting is restricted to the 002 prefix, 35,546 regular-season games.

Player-level and matchup features were built and gated rather than skipped. The corpus holds a who-guarded-whom matchups cache, defended-FG% by shot category, zone defensive RAPM, hustle counts, ten Synergy play-types, a 10-dimension player skill posterior and 151,914 lineup stints. Tested against the shipped stack, the defensive-RAPM team aggregate returns +0.0011 with an interval crossing zero and the defender-aware variant +0.0037 not significant. Composition built from RAPM sums scores 0.608 against the shipped DARKO version. Physical head-to-head attributes are absent entirely: matchup geometry needs commercial tracking data the project does not hold. The one player-level term that survived is the scalar availability composition above.

Accuracy. Normalized gap is $(ll_{us} - ll_{mkt}) / (ln2 - ll_{mkt})$, the share of the closing line's skill-above-a-coinflip the model fails to capture; zero is parity. On the fully-covered frame (2019-20 $\rightarrow$ 2025-26, K=7, n=8,286) the market-blind model sits 13.59% behind the market closing line, and the closing line beats it in every season of the frame.



Log loss by season, all four forecasts on identical games, each converted to a probability with its own walk-forward scale. Lower is better. The closing line is below everything else in six of seven seasons; the market-blind model is above the opening line in six of seven. Series are distinguished by line style as well as colour.

season	opening line	closing line	offset construction	market-blind model
2019-20 (COVID)	0.6102	0.6096	0.6082	0.6158
2020-21 (COVID)	0.6257	0.6174	0.6266	0.6376
2021-22	0.6170	0.6039	0.6153	0.6220
2022-23	0.6326	0.6239	0.6304	0.6359
2023-24	0.5923	0.5825	0.5911	0.5993
2024-25	0.6040	0.5809	0.5947	0.5915
2025-26	0.5801	0.5727	0.5786	0.5875
pooled (n=8,286)	0.6084	0.5980	0.6059	0.6122

The market-blind model is the only one of the four above the opening line. It does not beat the price it would transact at, and it is beaten decisively by the close. Expressed as the share of open-to-close information recovered, the offset construction captures +24.1% and the market-blind model -36.4%, moving the wrong way. That result is the reason the strategy is built as a correction to the line rather than as a forecast compared against it. Measured against the closing line alone the normalized gap is 13.59%, and the two COVID seasons are the extremes in both directions — 2019-20 the model's best on this frame and 2020-21 its worst. Both are kept because they are fully injury-covered, which is the frame's only criterion.

Two further diagnostics, both on all games rather than the selected book. Against the opening spread the market-blind margin covers 50.65%, above a coin flip and significant, and below the 52.38% a -110 book charges, also significant. We disagree with the opening line by 2.455 points on average; if that disagreement were entirely real information we would cover 57.6%, so the genuine content is 0.206 points, 8.4% of what we claim — close to the third of face value the offset ridge independently arrives at. Closing-line value is positive and significant across 19 seasons, so the disagreement does carry information relative to the market's final price, by a margin smaller than the cost of transacting it.

Considerations

Reporting frames

Two frames, each defined by what data exists. **Model accuracy:** 2019-20 → 2025-26 — the daily injury report the availability leg is built on begins 2018-12-17, so 2019-20 is the first fully covered season. **Betting:** headline on 2023-24 → 2025-26, the era of the measured multi-book panel, with the full seven-season frame reported beside it in the same table.

Execution

Model held fixed and only the transacted price varied, on the 2023-26 book.

execution	2023-26 ROI	status
1 retail book	+10.63%	degenerate reference
2 books	+12.91%	measured in 2023-24 only
5 books	+15.31%	firm baseline
9 books	+16.62%	max books observed, the reported tier
exchange, 2% commission	+14.72%	arithmetic, no exchange data held

The ladder flattens fast: 36% of the time two books post the same number, so extra books duplicate rather than add. The exchange row reprices executed-elsewhere bets under an assumed commission and is not a measurement.

Methodology

- **Walk-forward, no lookahead.** Configuration chosen on seasons 1..k from a pre-declared space, scored on k+1 only.
- **Settlement-only fills.** Priced at the recorded opening spread, held to settlement at -110. No queue model, no partial fill, no re-pricing.
- **Bet-time information only.** Availability is built from the last injury report published *before* game day, never the 5PM report or the pregame inactive list, both of which post after the opening line.
- **Regular season only.** Filtered on the 002 prefix. NBA Cup group-stage games are in because they count in the standings; a pre-registered difference-in-differences found no detectable effect (MDE 2.21 points).
- **Season-clustered inference,** checked across rolling-origin, leave-one-season-out, block-bootstrap and legacy splits, corrected for multiple comparisons across the running family.

Caveats

- **Simulated.** The model is not in production and no capital has been deployed.
- **Best-of-9 pricing is observed in one season and inferred in two.** A measured multi-book panel exists for 2023-24 only, at 7.74 books per game; 2024-25 observes 1.00 and 2025-26 observes 1.03, so their price comes from a shopping law applied to one recorded book.
- **Best-of-N transacts at whichever book is furthest offside.** 8.1% of best prices sit more than 1.5 points off the next book — the ones that get limited, lowered or voided. Charging for them costs about 2.3 ROI points, and that charge captures none of the staking limits or account restrictions that arrive faster in practice.
- **2026-27 is the first genuinely out-of-sample season.** The betting configuration is re-selected walk-forward, but the model architecture was chosen on a 2021-26 corpus and handed to every step as fixed.
- Exchange fees, state taxes and account limits are not modelled. Full method, gates and the register of rejected work are in the repository.

Next steps

- **Capture at least two books at the open, from opening night.** Best-of-two lifts CLV by about 49%, and taking the worse book erases nearly all of it. The logger has never run in-season, so if it is not up on opening night the season's open-price CLV record is lost and cannot be reconstructed. Free, and the highest-value operational change available.
- **Run both arms prospectively in 2026-27.** The offset construction is the primary and the market-blind model is its live control; they share 65% of their bets and take the same side on every shared game, so this is a controlled comparison rather than a portfolio.
- **CLV as the live yardstick.** Significant across 19 seasons and positive in 17, in units that need no devig convention, and it resolves in weeks where ROI needs decades.
- **Exchange access, if it becomes available.** The largest single lever measured here and an access problem rather than a modelling one.
- **Props before sides.** Both shipped improvements of the last cycle came from the props engine, and soft books are softer on props.

Not next: further feature search on the sides model. The question the project turns on is whether live closing-line value against opening prices, with two books captured, behaves as the backtest says.